

## Tut 4 Answer key

1. a.  $i(0^+) = 1 \text{ A}$        $v(0^+) = 2 \text{ V}$   
b.  $\frac{di(0^+)}{dt} = 25 \text{ A/s}$        $\frac{dv(0^+)}{dt} = 0 \text{ V/s}$   
c.  $i(\infty) = 6 \text{ A}$        $v(\infty) = 12 \text{ V}$
2.  $V(t) = 5.33(e^{-2t} - e^{-0.5t}) ; t > 0$
3. a.  $\alpha = 5 \times 10^8$        $\omega_0 = 10^9$   
b.  $s_1, s_2 = -5 \times 10^8 \pm j8.66 \times 10^8$   
c.  $i(t) = e^{-5 \times 10^8 t} (A_1 \cos(8.66 \times 10^8 t) + A_2 \sin(8.66 \times 10^8 t))$
4.  $i(t) = 15e^{-2t}(\cos 2t + \sin 2t) ; t > 0$
5.  $v(t) = 5 \left(1 - e^{\frac{-200t}{9}}\right) ; t > 0$        $i(t) = \frac{10^{-3}}{3} \left(e^{\frac{-200t}{9}}\right) ; t > 0$
6. a.  $i(t) = 5e^{-0.5t} ; t > 0$       b.  $i(t) = 6e^{\frac{-2}{3}t} ; t > 0$
7. a.  $E_L = 12.5 \text{ J}$  and  $E_C = 56.25 \text{ J}$   
b.  $i_L(0^+) = 5 \text{ A}$ ,  $V_C(0^+) = 15 \text{ V}$ ,  $i(0^+) = 0$ ,  $i_C(0^+) = -5 \text{ A}$   
c.  $i(t) = 3 - e^{-2t}(3 \cos t + \sin t)$  or  $i(t) = 3 - 3.16e^{-2t}(\sin t + 71.56^\circ)$
8. a.  $v(\infty) = IR_1$   
b.  $v(0^-) = 0$   
c.  $v(t) = IR_1 \left(1 - e^{\frac{-t}{(R_1+R_2)C}}\right)$